

SGA 2: Application to the Fundamental Group

Bounded source-aligned working-unit successor — 22 July 2026

Authority and boundary. Corrected French TeX lines 3544–3553; original printed p. 120, source-PDF physical p. 104, and recomposed running p. 96. The transition on line 3542 and blank line 3543 are excluded. Blank line 3554 and the proof beginning on line 3555 are excluded. The raw continuation cursor is line 3554 and the next substantive cursor is line 3555. The French authority remains byte-identical. The same-edition PDF is manifestation and layout evidence only; the external English candidate is comparison only. The line 3551 source grammar defect is disclosed under its final manager-adjudicated stable ID. Independent review and archive handoff have not been performed; this package is internal and not for release.

Lemma 3.9. — *Let A be a noetherian local ring, and let $t \in \mathfrak{r}(A)$ be an A -regular element. Suppose that A is complete for the t -adic topology and is moreover a quotient of a regular local ring (for example, if A is complete). Set $B = A/tA$.*

- (i) *If, for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$, one has $\text{depth } A_{\mathfrak{p}} \geq 2$, then the purity of B implies the purity of A .*
- (ii) *If, for every prime ideal $\mathfrak{p}$ of A such that $\dim A/\mathfrak{p} = 1$, one has $\text{depth } A_{\mathfrak{p}} \geq 2$, if $A_{\mathfrak{p}}$ is pure whenever $t \notin \mathfrak{p}$, and if⁽⁵⁾ $\text{depth } A_{\mathfrak{p}} \geq 3$ whenever $t \in \mathfrak{p}$, then the purity of A implies the purity of B .*

Source note [SGA2-X-L3551-MISSING-EST-SRCDEF-001]. The French authority prints *si $A_{\mathfrak{p}}$ pur lorsque $t \notin \mathfrak{p}$* , omitting the finite copula *est*. The English supplies “is” in the unambiguous intended condition. The French authority remains unchanged.

⁽⁵⁾N.D.E.: This last condition can be improved; cf. the editor’s note (1) on p. 90.